Version: V1.0 Revision Record

Date	Revision Content	Version	Reviser	Reviewer

Disclaimer and Limitation of Liability

- This manual is designed for the use of the OHW808 product. Without the written consent, no company or individual is allowed to reproduce and back up this manual in any form (electronic, mechanical, photocopying, recording, or other means).
- All information, specifications, and illustrations in the manual are the latest as of the publication deadline. This company reserves the right to make changes without prior notice. Although the content of the manual has been carefully reviewed, its completeness and accuracy (including but not limited to product specifications, functions, and illustrations) are not guaranteed.
- This manual is for the use of professional automotive repair technicians only.
- This manual only provides the operating methods for OHW808 products. Our company assumes no responsibility for any consequences resulting from the use of these operating methods on other equipment.
- This company shall not be liable for any costs and expenses resulting from accidental incidents caused by the user or a third party; or damage, loss of equipment, and other issues caused by the user's misuse, abuse, unauthorized modification, repair, or failure to operate and maintain the equipment in accordance with the requirements of the manual.

Maintenance and Usage Precautions for OHW808 Main Units

- Do not disassemble this unit without authorization;
- Avoid subjecting this unit to severe impacts;
- Keep this unit away from magnetic fields;
- Do not keep this unit in a high-temperature environment for a long time;
- Do not keep this unit in a low-temperature environment for a long time;
- Do not press the screen too hard or use sharp objects when touching it;
- Do not clean this unit with water and chemical solvents. It should be cleaned with a soft cloth and neutral cleaner.

Car Inspection Precautions

- Operations must be carried out in accordance with the safety rules of the automotive repair industry. Pay special attention to the impact or damage caused by environmental factors such as surrounding acids, alkalis, toxic gases, and high - pressure heavy objects;
- The liquid in a car's battery contains sulfuric acid, which is corrosive to the skin. During operation, direct contact between battery fluid and the skin should be avoided. Particular attention should be paid to preventing it from splashing into the eyes, and open flames must be kept away;
- The exhaust gases emitted by the engine contain a variety of toxic compounds. Inhalation should be avoided, and the vehicle should be parked in a well - ventilated area during operation;

- temperature components such as the radiator and exhaust pipe should be avoided;
- • Before starting the engine, the parking brake should be engaged, and the gear lever should be placed in “Neutral” (for manual transmissions) or “Park” (for automatic transmissions) to avoid accidents when the engine is started;
- Before repairing the vehicle, engage the parking brake, shift the transmission into Neutral or Park, and lower the driver's - side window;
- If the engine can be started, warm it up to a normal temperature (with the water temperature around 80℃) and turn off auxiliary electrical appliances (such as air conditioning, lights, and audio);
- Locate the diagnostic socket of the vehicle, and check to confirm that the socket's wiring is in good condition. Then connect the main unit for diagnosis. Otherwise, do not conduct the test to avoid damaging the main unit. If necessary, use a multimeter to measure the voltage of the diagnostic socket.

Precautions for Instrument Use

- When using the OHW808 product for testing, it must be handled with care, kept away from heat sources and electromagnetic fields to avoid interference with the main unit;
- Do not use sharp objects to touch the screen. It is recommended to use the matching stylus for operation;
- Do not disconnect the circuit when electrical components are powered, to prevent self - induced and mutual - induced currents from damaging sensors and the car's ECU (Electronic Control Unit);
- When electrical appliances are working normally, it is strictly forbidden to bring magnetic objects close to the car's control unit, as

it may cause damage to the control unit;

- When removing or installing the car's control unit or electrical components, it must be done 1 minute after the ignition switch is turned off;
- Never operate the diagnostic device while driving the vehicle to avoid causing a traffic accident.

Table of Contents

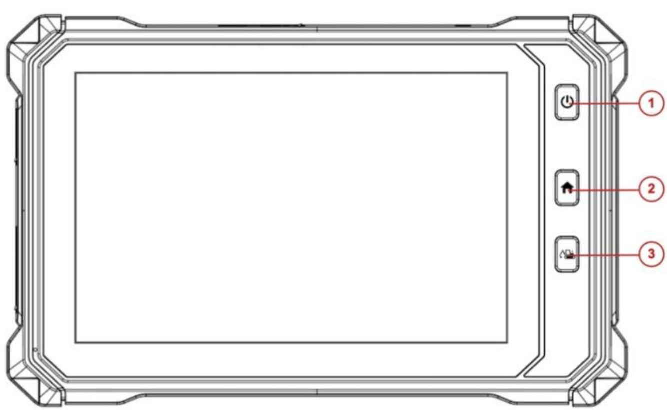
1 Introduction to Product Structure and Accessories

1.1 Product Overview

The OHW808 Car Fault Diagnostic Tool is a comprehensive automotive computer fault diagnostic tool. It is for the detection and diagnosis of electronic control systems in diesel and other vehicles. The product is suitable for small, medium, and large repair enterprises, training institutions, car manufacturers, repair stations, diesel engine manufacturers, mining machinery, petrochemical energy companies, and so on.

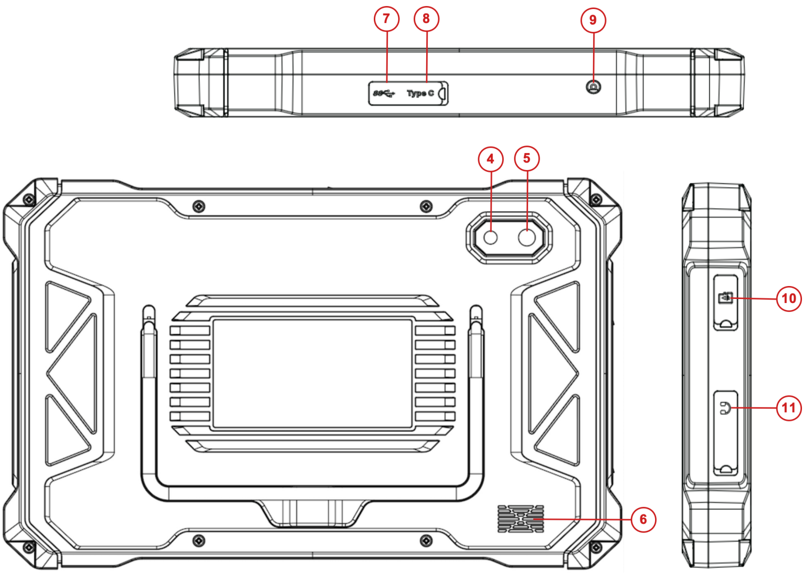
The product's software configuration is comprehensive, and the vehicle data and information are authoritative. It fully meets the strict requirements of customers for the breadth and depth of detection. The software covers data for abundant of models, both domestic and imported, and provides a powerful repair information assistance system. This enables users to quickly and conveniently solve problems in their practical work, thereby improving work efficiency and demonstrating the advantage of professional standards.

1.2 Product Structure Description



Serial number	Name	Description
①	Power switch	Used to switch on or off host, or lock screen

②	Home	Return to App Main Menu
③	Diagnostic	Enter the main diagnostic interface

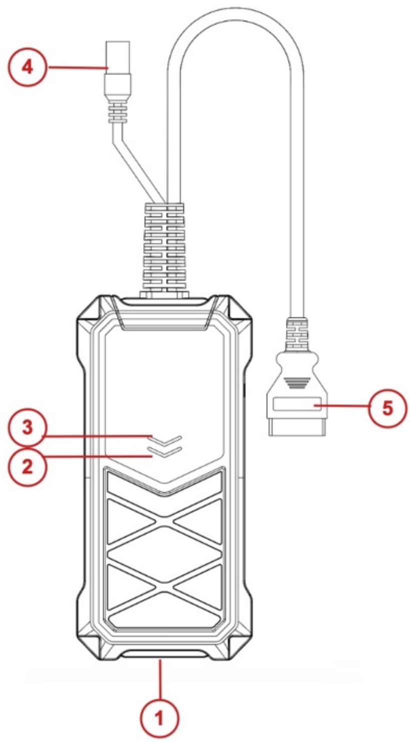


Serial number	Name	Description
④	Camera	Used to take a photo or record a video
⑤	Flashing light	Used to provide light when the light is weak
⑥	External speaker port	For external sound playback
⑦	USB port	Connect host to the VCI or connect to the USB flash drive.
⑧	USB type C	Charge or transmit data
⑨	DC power port	Charge（DC 12V,2A）
⑩	TF card slot	Storage TF card location（Optional）
⑪	3.5 headphone jack	headphone jack

Host parameter

Name	Specifications
CPU	Quad-core1.8GHz processor
System	Android 11
Memory	8GB RAM &128GB ROM
Screen	10.1” 16:10, 1280*800
TP touch screen	Touch capacitive screen
Camera	800 million pixels, with flashing light
Wi-Fi	2.4G+5G
Bluetooth	BT4.2/5.0
Sensor	Light Sensor/G sensor
Battery	10000mAh/3.7V
Size	306*197*26mm
Work temperature	-10~50℃
Storage temperature	-20~60℃

1.3 VCI Box Structure Description



Serial No.	Name	Description
①	USB Type-C	Connect with tablet host or upgrade
②	Diagnostic indicator	Light flashes when communicating with the vehicle
③	Power indicator	Light on when the power supplied
④	DC Power Port	Charge by DC power supply
⑤	OBD Diagnostic Port	Connect the vehicle

Main Parameter

Processor	32Bit RISC-V
Frequency	648MHZ
Bluetooth	BT4.2(BLE)
RAM	800KB
Flash	8M

2 Powering On and Off the Host Device

2.1 Power Supply for the Host

The host can be charged in the following ways:

- AC/DC power adapter
- Built-in battery pack

Connect one end of the AC/DC power adapter to the DC power port of the host, and the other end to a wall socket. The power adapter can charge the built-in battery pack.

Note! The voltage of the power supply should be within the applicable range of the host product. Exceeding the range may cause damage to the product.

2.2 Powering On

Press and hold the power switch of the host (for about 3 seconds) to power it on. The host will enter the following startup interface, and the system will begin to run.

2.3 Powering Off

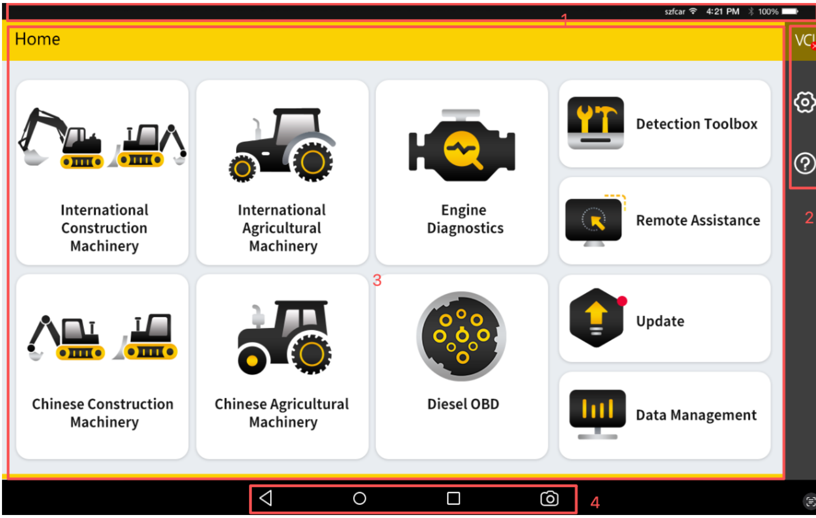
Before powering off, please first stop all measurement items and disconnect the probes from the test vehicle and oscilloscope, then click 【◁】 to exit the program.

The steps to shut down the tablet host are as follows:

1) Press the power switch of the host for a short time (about 3 seconds);
2) Select [Shut Down] in the pop-up prompt;
3) Click [OK] to turn off the host.

2.4 Introduction to Various Function Menus

After the system is powered on, the following main function menu will be entered. The main function menu of each version and model will be slightly different. Please take the display of the model you purchased as the standard. The following figure is for the diesel version:



1) Status Icons: These are the default icons of the standard Android operating system
2) Toolbar (See Table 1)
3) Main Function Menu (See Table 2)
4) Navigation Bar (See Table 3)

Tip: It is recommended to lock the screen at any time when the device is not in use to protect your system information security and save battery power. Gently press the power/lock screen button, and the screen will automatically lock. Pressing too hard or holding the button for a long time may cause the button to malfunction or enter the shutdown interface.

Table 1：Toolbar

Icon	Function Name	Function Description
	VCI Connection	VCI Box Connection and Status Display (Available throughout the entire diagnostic operation)
	Screenshot	One-Click Screenshot of the Current Visible Screen (Available throughout the entire diagnostic operation)
	Setting	Shortcut Key for "Settings" Function

Table 2：Main Function Menu

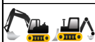
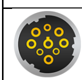
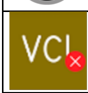
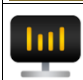
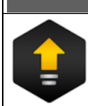
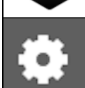
Icon	Function Name	Function Description
	International Construction Machinery	International Construction Machinery Diagnosis
	International Agricultural Machinery	International Agricultural Machinery Diagnosis
	Engine Diagnostics	Engine System Diagnosis
	Chinese ConstructionMachinery	Chinese Construction Machinery Diagnosis
	Chinese Agricultural Machinery	Chinese Agricultural Machinery Diagnosis
	HD OBD	Diesel OBD Diagnosis
	VCI Connection	Establish and manage communication connection with VCI device, see Section 3.2
	Data Management	Used to browse and manage saved data files, see Section 8
	Remote Assistance	Run this program to establish remote assistance with after-sales technical team, see Section 9
	Detect Toolbox	Node Screening、Pin Detect
	Reference	Provide device usage instructions, repair help, fault code queries, and other help information, see Section 10
	Update	Online upgrade of system software, vehicle model software, etc., see Section 11
	Setting	Settings and viewing system information, see Section 12

Table 3: Navigation bar

Icon	Function Name	Function Description
	Volume	Turn Down Volume
	Back	Return to the previous interface
	Home	Return to the Android system main interface
	Recently Used Programs	Display a thumbnail list of recently used programs. Clicking on a program thumbnail will open that program, and swiping up on a program thumbnail will close that program
	Volume	Turn Up Volume
	Screenshot Tool	Take a screenshot of the current screen

3 Getting Started

The diagnostic program establishes a data connection with the vehicle's electronic control system through the connected VCI box device, enabling it to read vehicle diagnostic information, view data streams, and perform actuation tests.。

➢ To establish good communication between the diagnostic program and the vehicle, the following operations need to be performed:

1) Connect the VCI box to the vehicle's diagnostic socket and provide power;
2) Establish communication between the VCI box and the OHW808 host via a USB data cable;
3) Check the VCI connection status in the top right corner of the screen (see Section 3.2.2). Once the connection is successful, vehicle diagnostics can be carried out.

➢ How to perform vehicle diagnostics:

1) Establish good communication between the diagnostic program and the vehicle under test, see Section 3.2
2) Select the diagnostic system, see Section 4.1

A detailed operation description will be provided below.

3.1 Technical Requirements Before Diagnosis

3.1.1 Equipment Requirements

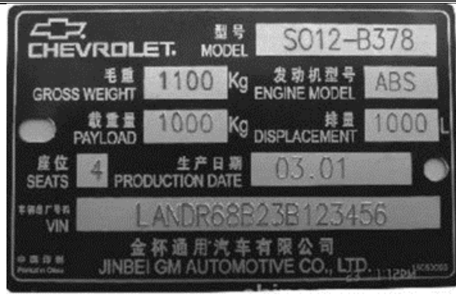
The OHW808 automotive computer fault diagnostic tool is equipped with a host and various test connectors upon leaving the factory. When testing, select the corresponding test connector according to the type of the vehicle's diagnostic socket.

3.1.2 Vehicle Requirements

- The ignition switch should be in the ON position;
- The car battery voltage should be between 11~14V or 24~27V (depending on the vehicle's power supply)
- The throttle pedal should be in the closed position, that is, the idle speed engagement point;
- The ignition timing and idle speed value should be within the standard range, and the water temperature and transmission oil temperature should reach the normal working temperature (water temperature 90~110℃, transmission oil temperature 50~80℃);
- The diagnostic circuit connection should be normal.

3.1.3 Technician Requirements

- Must have basic knowledge of automotive electronics;
- Understand the basic operation methods of this product and have read this manual thoroughly;
- Be able to basically distinguish whether the fault phenomenon of the vehicle being tested belongs to mechanical fault or electronic control part fault;
- Be aware of the vehicle's origin, production year, model, engine type, and other information.



3.2 Vehicle Connection

3.2.1 Connection of VCI Box to the Vehicle

Before VCI box connected to vehicle, it is necessary to judge whether the diagnostic socket of the test vehicle is a standard OBD-II port or a non-standard OBD-II port.

- For Vehicles compatible with the OBD-II management system, The VCI can be connected to vehicle diagnostics socket and supplied with power only with the integrated standard OBD-II connector;
- For Vehicles that are not compatible with the OBD-II management system, you need to select the corresponding connector; some other vehicles need to supply power to the VCI box through other power sources of the vehicle.

Here we make the operating instruction regards to these two connection modes.

➢ **Standard OBD-II port connection**

For vehicles with standard OBD-II port, you just need to connect with all-in-one main test cable OBD connector rather than other connectors, as shown in Figure 3.2-1:

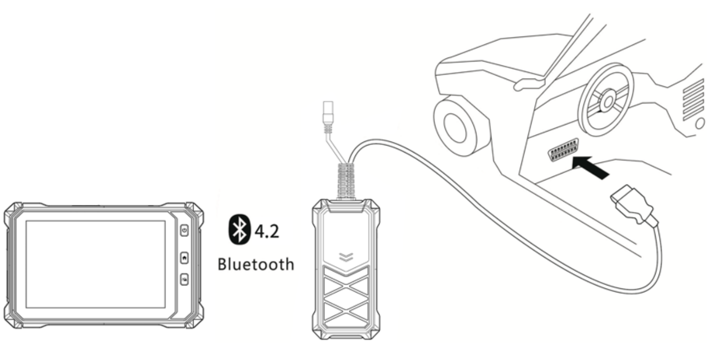


Figure 3.2-1 Connection of standard-OBD-II port

Instructions:

1) Determine the location and the port of the diagnostic socket;
2) Connect the integrated OBD-II port of the VCI to the vehicle diagnostic socket;
3) At this time, the VCI box is powered by the vehicle diagnostic socket, and the power indicator light is on.

Note: After test is completed, please rotate the fixing bolts and then gently unplug the main test cable to avoid damage to the diagnostic port.

- **NON-OBD-II PORT CONNECTION**

For vehicles connected to non-OBD-II port, you need to connect the main test cable to their corresponding dedicated connectors, as shown in Figure 3.2-2:

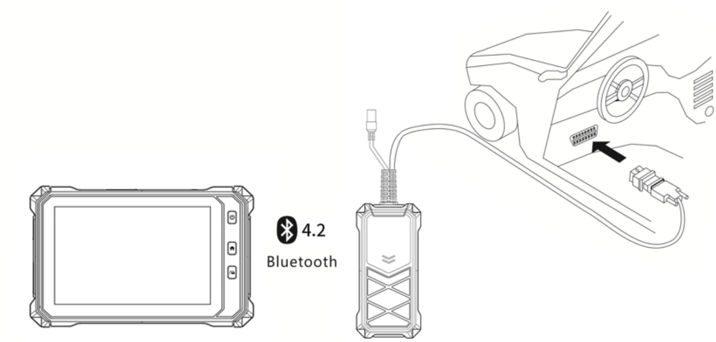


Figure 3.2-2 Connection of non-OBD-II port

Instructions:

1) Determine the diagnostic socket location and port, and whether need the external power source;
2) Connect the main test cable of the VCI to a dedicated adaptor corresponding to the vehicle;
3) Connect the dedicated connector that is connected to the main test cable to the vehicle diagnostic socket;

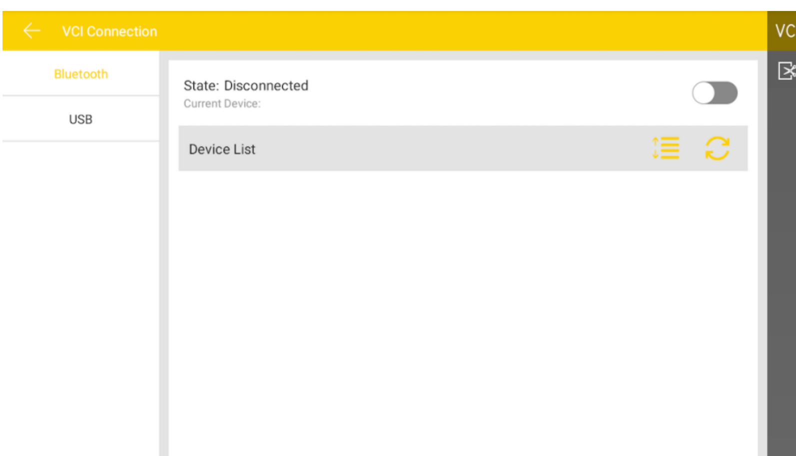
At this time, VCI box is powered by the vehicle diagnostic socket, and then power indicator light is on (if it is not lit, it may be because the vehicle diagnostic socket is not energized, you can energize the VCI box by the cigarette lighter or the battery clip).

3.2.2 Connection between Host and VCI Box

To perform vehicle diagnostics, the host and VCI box need to be connected and matched. The VCI box supports two connection methods with the OHW808 host: Bluetooth and USB cable.

Pairing via Bluetooth:

1) Turn on the power of the OHW808 host;
2) In the main function menu, select [VCI Connection]; choose [Bluetooth] as the connection method;
3) Click the scan  ”icon on the right side of the device list, and the system will automatically scan nearby Bluetooth devices;
4) Select the target Bluetooth device for pairing;
5) After the pairing is completed, the VCI icon in the top right corner of the screen will change from  ”to ”, indicating that the Bluetooth pairing is successful and vehicle diagnostics can be carried out.



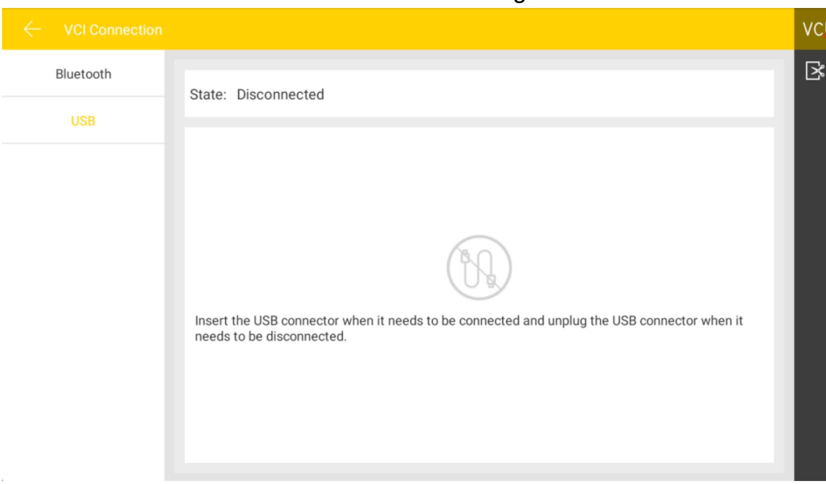
Note: If the Bluetooth device cannot be found, it may be that the signal strength of the transmitter is too weak. In this case, please get as close as possible to the VCI Bluetooth device.

Connection via USB Cable:

USB connection is the fastest communication method between the OHW808 host and the VCI box. Please use the dedicated USB cable provided by our company for connection. After the connection is completed, the VCI icon in the top right corner of the screen will change from 

"to " 

",indicating that the USB connection is successful and vehicle diagnostics can be carried out.



4 Automotive Diagnosis

The OHW808 automotive diagnosis is categorized based on the origin, purpose, and system of the vehicles: International Construction Machinery Diagnosis、International Agricultural Machinery Diagnosis、Engine System Diagnosis、Chinese Construction Machinery Diagnosis、Chinese Agricultural Machinery Diagnosis、Diesel OBD Diagnosis、Others。The vehicle model selection is menu-guided. The technician can make a series of choices according to the actual configuration of the vehicle to be tested, following the on-screen prompts, until the vehicle to be tested is accurately identified. The specific selection items will vary depending on the vehicle model being tested.

After the vehicle is connected (for details, see Section 3.2), click on [Vehicle Diagnosis] in the main menu to start the vehicle diagnosis, as shown in the figure below:

1) Toolbar, see Table 1 for details;

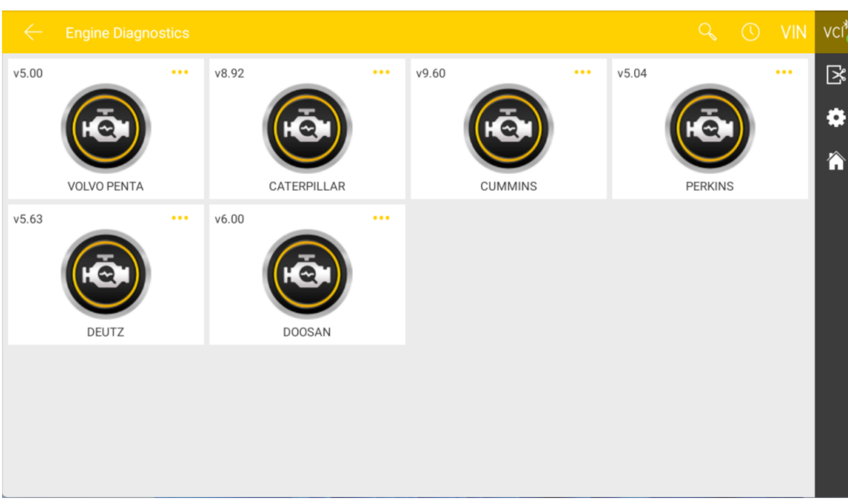


Table 1:

Icon	Function Name	Function Description
	Query	Query diagnostic vehicle models
	History	View current packet version and historical version
	VIN	Automatic VIN Scanning
	VCI Connection	VCI box connection and status display (available throughout the entire diagnostic operation)
	Screenshot	O ne-click screenshot of the current visible screen (available throughout the entire diagnostic operation)
	Settings	Shortcut key for the "Settings" function
	Home	Return to the main menu

4.1 Vehicle Model Selection

The diagnostic program needs to select the vehicle model before entering the system module diagnostic function. The vehicle can be identified in the following ways:

- By manual selection
- By automatic identification
- By OBD Ⅱ access mode

4.1.1 Manual Selection

Manual vehicle model selection is a menu-guided mode. You only need to follow the screen prompts and make a series of selections. The specific options will vary depending on the vehicle model being tested. You can also quickly find the vehicle brand through the search function in the toolbar, and then make selections according to the screen prompts.

4.1.2 Automatic Identification Selection

The latest "auto-recognition" feature of the OHW808 intelligent diagnostic system can skip the manual selection step. It directly obtains specific vehicle information from the vehicle's ECU and quickly enters the diagnostic interface after confirmation. Currently, this function supports some vehicle models, and the specific models are subject to the function menu display. Taking the [Bosch system] as an example:

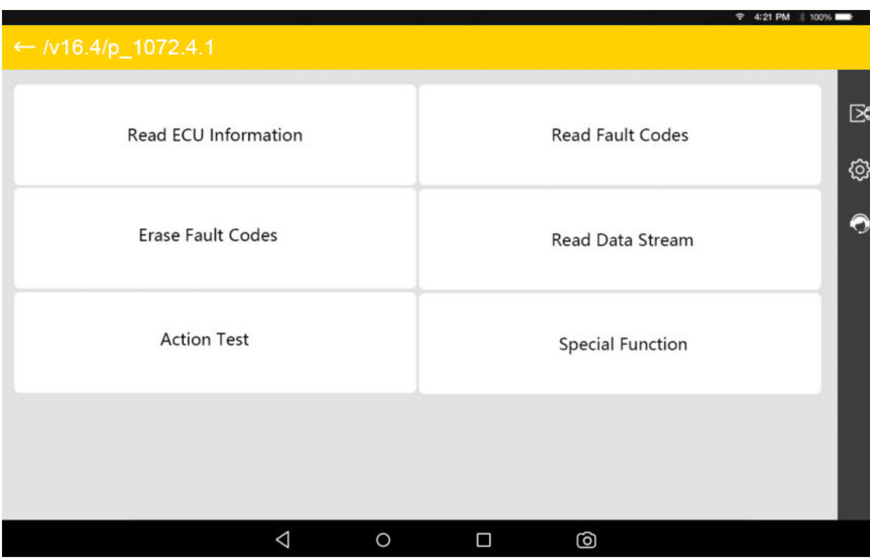
1. In the commercial vehicle diagnosis selection interface, choose [Electronic Control System];
2. In the electronic control system interface, click on [Bosch - Bosch (China IV, V, VI Auto-Recognition)] to automatically recognize vehicle information.

4.1.3 Generic OBD Access Mode

In some cases, due to the database not supporting or the vehicle having other functional characteristics, the diagnostic instrument may not be able to identify the vehicle and establish communication through normal channels. At this time, you can use the OBD direct access function to perform general OBD II or EOBD tests. For more details, see section 4.6.

4.2 Introduction to Diagnosis and Other Advanced Functions

After the vehicle model selection is completed, you can proceed with the vehicle diagnosis. The main diagnostic interface is shown in the figure below:



1) Toolbar and status bar (see Table 1 for details)

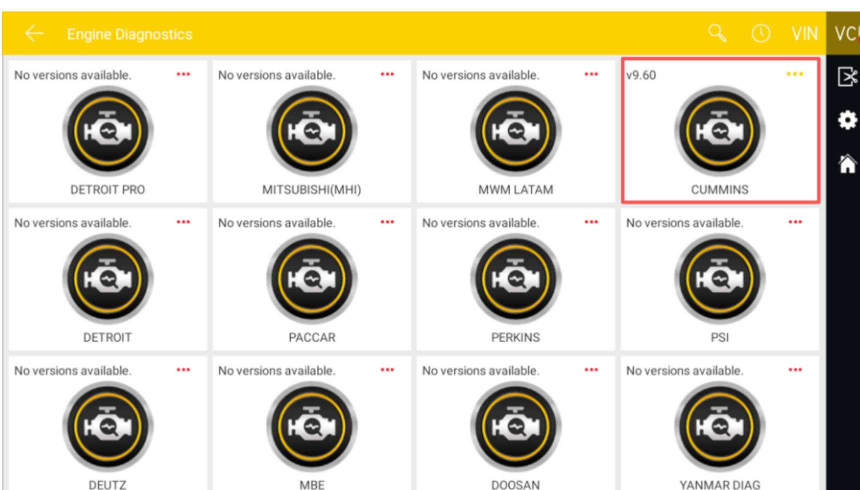
2) Main interface of diagnostic functions (see Section 4.3 for details)

Table 1：

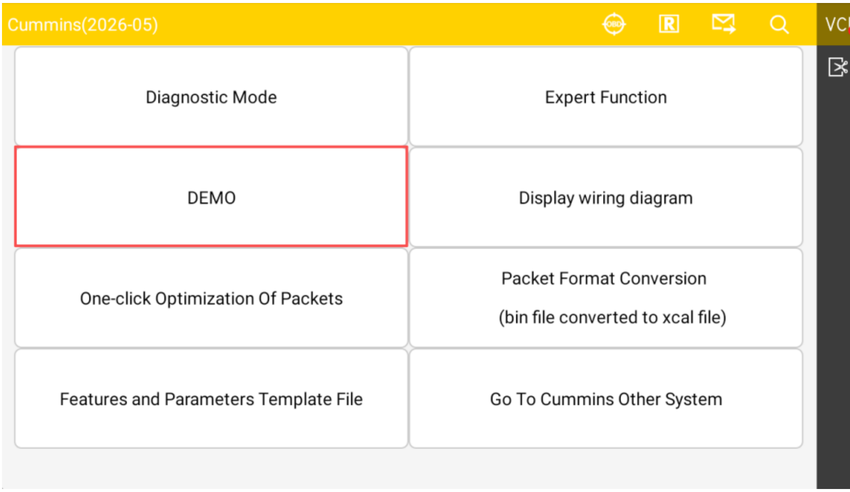
Icon	Function Description
	Vehicle OBD Location Diagram
	Create a test report, which can be viewed, printed, and sent in "Data Management"
	One-click feedback to report issues encountered during vehicle diagnosis to the FCAR after-sales technical team
	One-click search for various brand vehicle models

4.3 Diagnosis

1) Taking [Engine Diagnostic——CUMMINS] as an example: Select CUMMINS to enter the diagnostic interface.

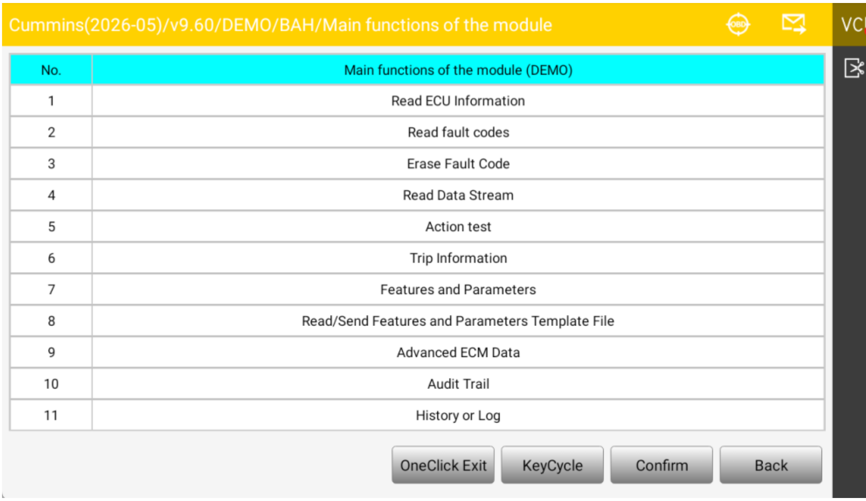


2) Select [DEMO];



3) Select [ISD4.5/6.7 CM2880];

4) Enter the main diagnostic page;



The main diagnostic interface usually includes the following options:

Read ECU Information： Read and display the control system module information detected from the ECU.

Read Fault Codes: Read and display the fault code information retrieved from the vehicle system module

Clear Fault Codes: Clear the fault codes and frozen frame data retrieved from the vehicle system module

Live Data： Read and display the real-time operating parameters of the current system module

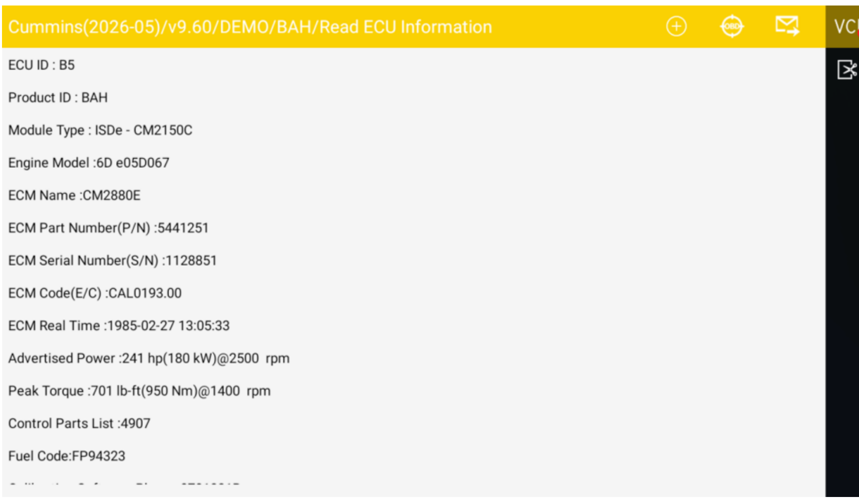
Actuation Test： Test the actuators of various vehicle systems. The test options may vary depending on the manufacturer and vehicle model.

Special Functions: Perform component adaptation or variable coding functions for customized configuration operations, and carry out adaptation for certain components after repair or replacement. The special functions vary with the selected vehicle model, and the displayed function names are slightly different.

Calibration Function: To facilitate repair and modification of engine power, the ECU has enabled calibration and fine-tuning functions.

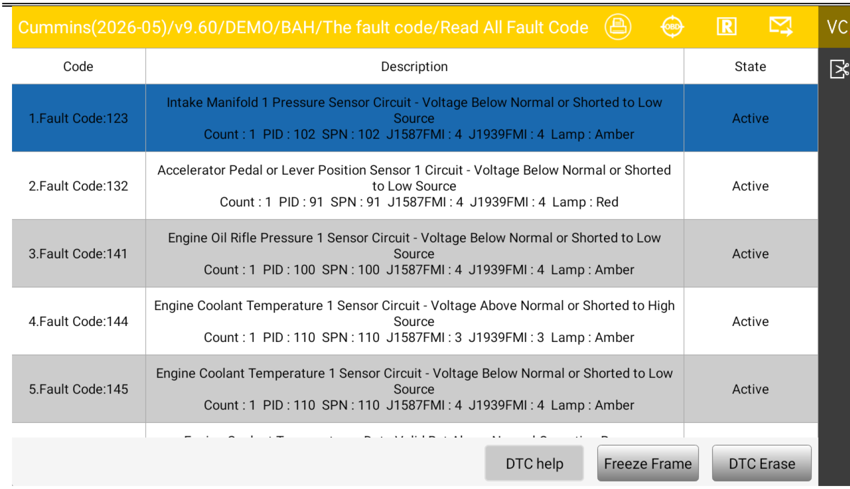
4.3.1 Read ECU Information

Read and display the ECU version information (hardware, software) retrieved from the vehicle under test.



4.3.2 Read Fault Codes

Read and display the fault codes retrieved from the system module of the vehicle under test and provide an explanation of the fault content;



Code	Description	State
1.Fault Code:123	Intake Manifold 1 Pressure Sensor Circuit - Voltage Below Normal or Shorted to Low Source Count : 1 PID : 102 SPN : 102 J1587FMI : 4 J1939FMI : 4 Lamp : Amber	Active
2.Fault Code:132	Accelerator Pedal or Lever Position Sensor 1 Circuit - Voltage Below Normal or Shorted to Low Source Count : 1 PID : 91 SPN : 91 J1587FMI : 4 J1939FMI : 4 Lamp : Red	Active
3.Fault Code:141	Engine Oil Rifle Pressure 1 Sensor Circuit - Voltage Below Normal or Shorted to Low Source Count : 1 PID : 100 SPN : 100 J1587FMI : 4 J1939FMI : 4 Lamp : Amber	Active
4.Fault Code:144	Engine Coolant Temperature 1 Sensor Circuit - Voltage Above Normal or Shorted to High Source Count : 1 PID : 110 SPN : 110 J1587FMI : 3 J1939FMI : 3 Lamp : Amber	Active
5.Fault Code:145	Engine Coolant Temperature 1 Sensor Circuit - Voltage Below Normal or Shorted to Low Source Count : 1 PID : 110 SPN : 110 J1587FMI : 4 J1939FMI : 4 Lamp : Amber	Active

4.3.3 Erase Fault Codes

Before clearing the fault codes, please record and save the retrieved fault codes for later review and comparison.

➢ How to Clear Fault Codes

1) In the diagnostic interface, select [Erase Fault Codes];

2) The diagnostic system will then successively pop up prompts saying "Turn off the engine and turn on the ignition switch" and "Fault codes and frozen frame data will be cleared. Do you want to continue?", select [Continue]; (Note: This step requires strict adherence to the menu prompts to operate the vehicle)

3) When another prompt message appears, select [Return], and re-execute the read fault codes function to verify whether the fault codes have been cleared.

➢ Fault Code Analysis Steps:

1) Read and record all fault codes;

2) Erase all fault codes;

3) Simulate the conditions that caused the fault and conduct a road test;

4) Read and record the fault codes at this time again;

5) Distinguish between sporadic fault codes (irrelevant fault codes or historical fault codes) and static fault codes (current fault codes or related fault

codes);

6) Distinguish between the primary fault codes related to the fault symptoms and the secondary fault codes;

7) Identify the primary fault code among many fault codes or related fault codes (it may be the cause of other fault codes);

8) Based on the above analysis, further precisely detect the status of the sensors, actuators, or control computers represented by the fault codes, as well as the exact location of the fault.

4.3.4 Live Data

By selecting this function, the screen will display the real-time operating parameters of the selected system module. The displayed data parameters may vary depending on the vehicle model or different control modules, as shown in Figure 4.3.4-1.

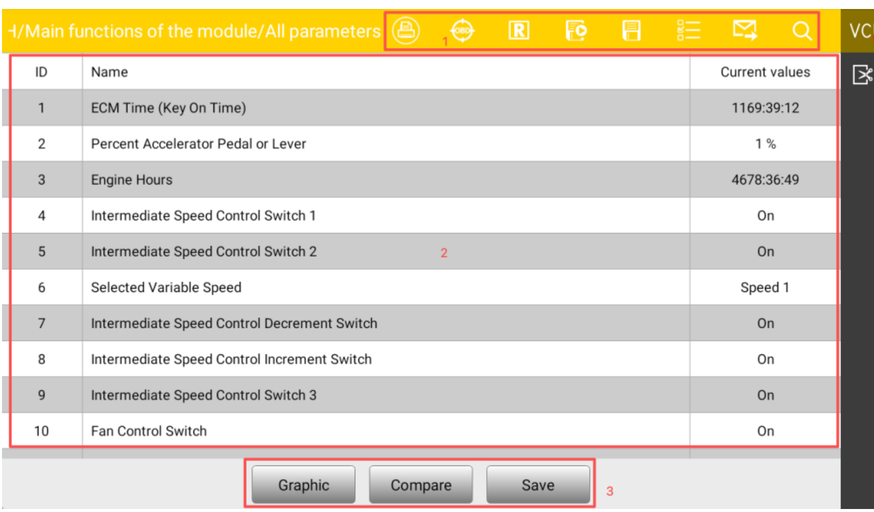


Figure 4.3.4-1 Data Stream Interface

1. Toolbar, see Table 1;
2. Data stream display area;
3. Data flow waveform graph, data flow comparison, data stream save.

Table 1：

Icon	Function Description
	Print Diagnostic Report
	Vehicle OBD Location Diagram
	Create a test report, which can be viewed, printed, and sent in "Data Management"
	Record data stream
	Save data stream/Excel table format
	Filter data stream
	One-click feedback, to report issues encountered during vehicle diagnosis to the FCAR after-sales technical team
	Search data stream (keywords)

➢ Data Flow Waveform Graph

In the data flow display interface, click the [Line Chart] function icon " " in the middle and lower part to view the data flow waveform graph of the operating parameters, as shown in Figure 4.3.4-2.

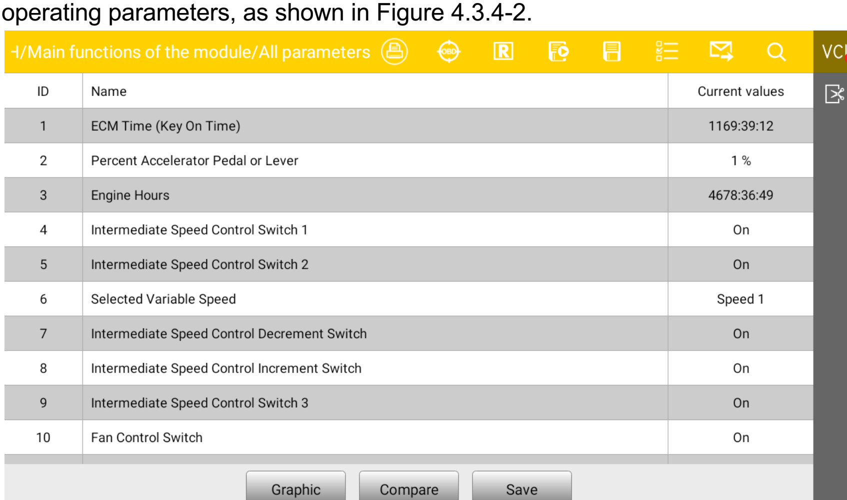


Figure 4.3.4-2 Line Chart

Data Stream Recording and Comparison Function

The OHW808 intelligent diagnostic system supports the recording and saving of data stream, and allows unlimited viewing in [Data Review]; it also supports the data flow comparison function. When the vehicle is in good condition, the data flow can be collected and saved to provide data reference for the next test or when encountering the same type of vehicle.

➢ How to Record Data Flow

1) Click the [Data Flow Filtering] function icon in the top right corner ”, select the parameters to be recorded in the pop-up window, and choose [OK] to return;

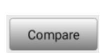
2) Select the [Data Flow Recording] function icon  ”, enter the file name to be saved in the pop-up window (it is best to name the file with vehicle information for easy reference), and choose [OK] to start recording;

3) The data flow starts recording. Slowly slide the touchscreen to the last data column; click the button in the bottom left corner 】 to stop recording and automatically save the data flow;

4) After the data flow recording is completed, the system will automatically save it (note: the data flow can be recorded for a maximum of 2 minutes), and the saved data flow can be viewed in [Data Review] under the Data Management function.

➢ How to Perform Data Flow Comparison

1) Click the [Data Flow Filtering] icon in the top right corner ”, select the parameters to be compared in the pop-up window, and choose [OK] to return;

2) Click the [Data Flow Compare] function icon“ 

”in the middle and lower part,select the saved data files of the same type of vehicle in the pop-up dialog box, and choose [OK] to start comparing;

3) At this time, a column of "Comparison Values" will appear in the data flow interface, as shown in Figure 4.3.4-3. The repair technician can quickly find and troubleshoot vehicle problems through the system parameter comparison.

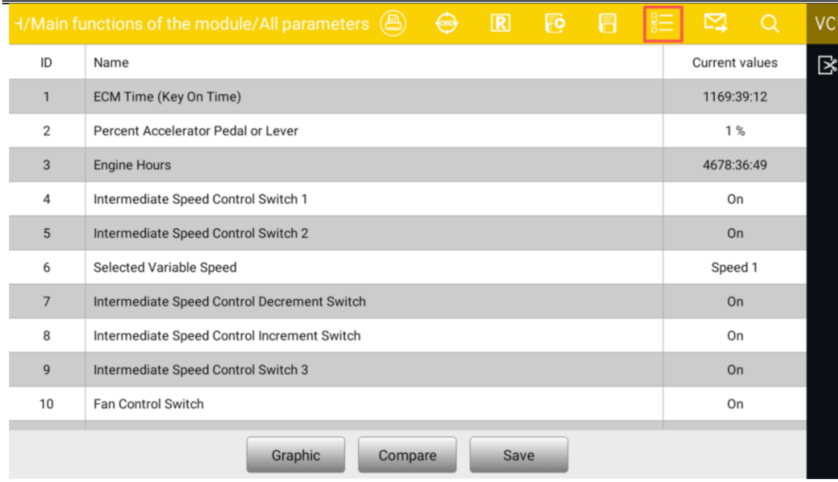


Figure 4.3.4-3 Data Flow Comparison

4.3.5 Actuation Test

This feature allows access to vehicle-specific subsystem tests and component tests. The available functions vary depending on the manufacturer, model year, and vehicle type, and the menu only displays the available test options.

When performing an actuation test, the diagnostic tool inputs commands to the ECU to drive the actuators, thereby determining whether the vehicle's electronic control system components and their circuits are functioning properly. Different vehicle models and control systems have different test options, so please refer to the menu display.

The following is an example of an actuation test for the urea pump directional valve in the Weichai engine system (Bosch Common Rail System).

- How to Perform Actuation Test
- Enter the diagnostic interface of the Engine Diagnostic——CUMMINS;
- Select [Action Test] in the diagnostic interface;
- Choose the test item [DPF Ash Reset];
- Confirm [Reset] to perform DPF Reset,Confirm [Reset] to perform DPF Reset as shown in Figure 4.3.5-1;
- Test whether the working status of the normal based on the execution situation, and click [Return] to exit the test.

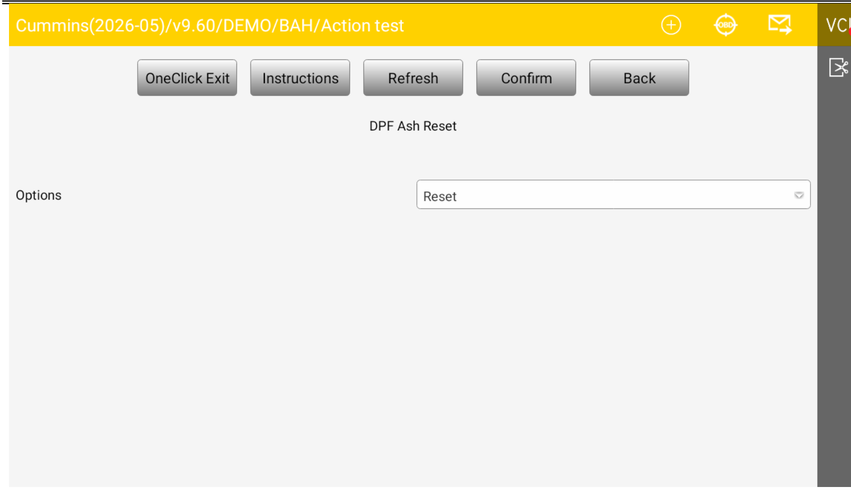


Figure 4.3.5-1 DPF Ash Reset

4.3.6 Special Functions

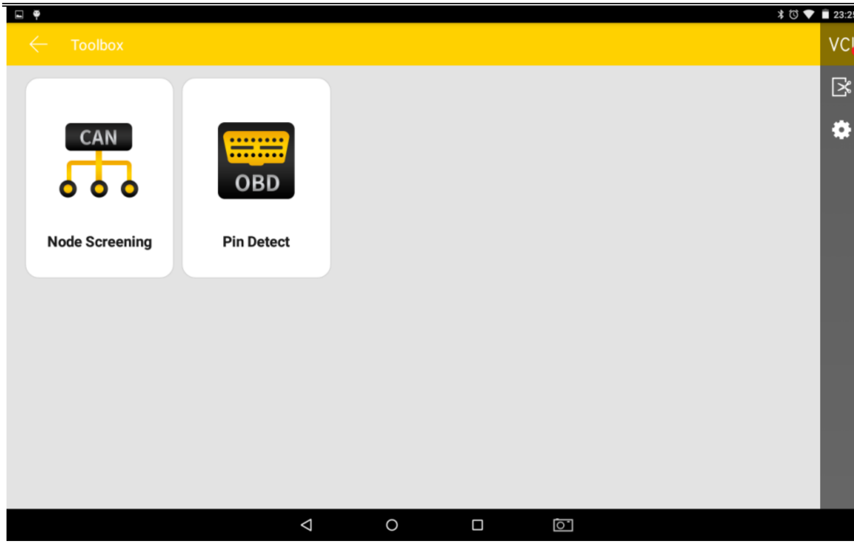
Special functions allow for adaptive operation of various components. They are mainly used to recalibrate or reconfigure components after maintenance or replacement, enabling mutual adaptation among the components of the electronic control system. Otherwise, the system will not be able to operate normally.

Carefully read the information on the screen and check the corresponding vehicle status, and strictly follow the menu prompts; select the appropriate execution item according to the instructions on the screen, enter the correct value or data, and then perform the operation.

After the operation is completed, the screen will display an execution status message such as "Operation Completed" or "Operation Successful".

5 Toolbox for Inspection

The OHW808 Automotive Intelligent Diagnostic Instrument, closely following market demands, has developed several commonly used auxiliary tools for repair technicians, including: CAN Node Screening and OBD Pin Detection, to meet daily repair needs.



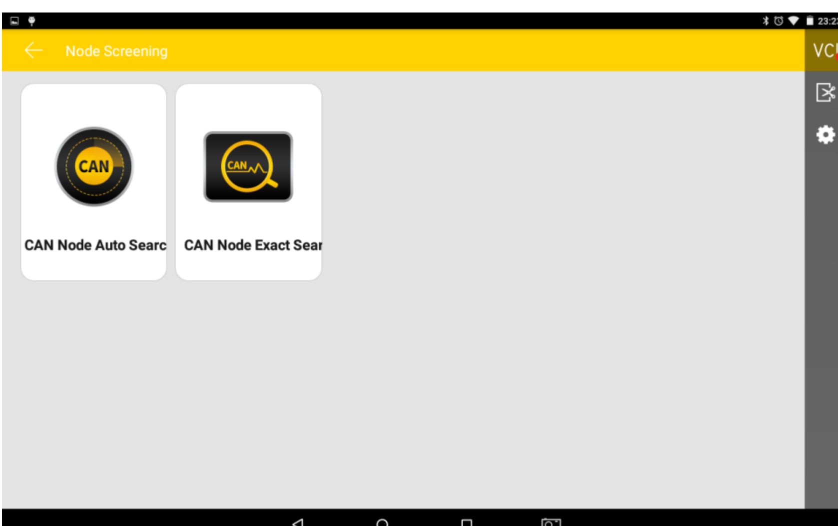
5.1 CAN Node Screening

After connecting to the vehicle's OBD interface (see Section 3.2), use this function to quickly determine the nodes that are communicating normally with the ECU via the CAN bus, in order to find the missing nodes.

How to Perform CAN Node Screening

Connect to the vehicle's OBD, see Section 3.2;

In the main menu, select [Inspection Toolbox] - [Node Screening] to enter the system;



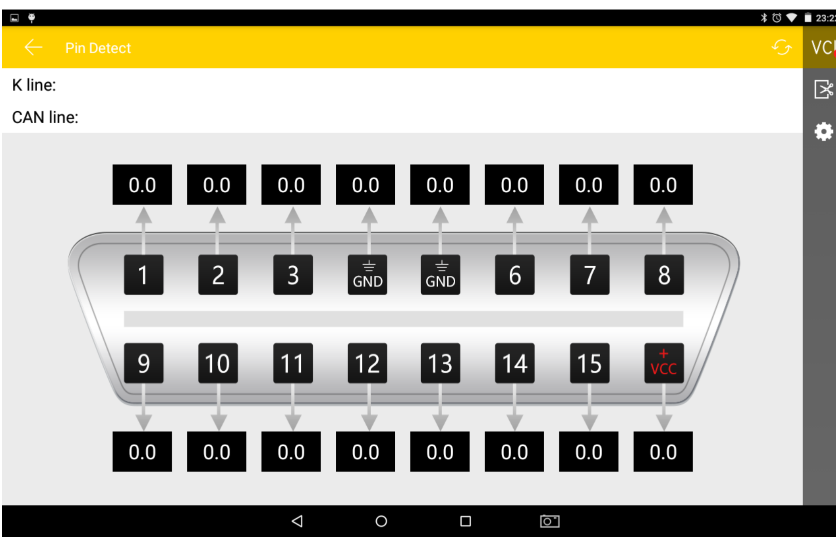
CAN Node Automatic Search: Select this method, and the program will automatically search and display the vehicle's CAN modules.

CAN Node Exact Search: Select this method to perform CAN node screening for specific systems.

5.2 Pin Inspection

This function is mainly for measuring the voltage of the 16-pin terminals of the OBD diagnostic interface, and also for determining the pin positions of the K-line and CAN-line.

After connecting to the vehicle's OBD interface (see Section 3.2), enter [Inspection Toolbox] and select [Pin Inspection]. The program will automatically measure the voltage of the 16-pin terminals of the OBD diagnostic interface and determine the pin positions of the K-line and CAN-line, as shown in the figure below:



6 Data Management

The "Data Management" application is used to save, view, or print saved files. Most of the files are generated by the toolbar operations in the vehicle model diagnosis interface.

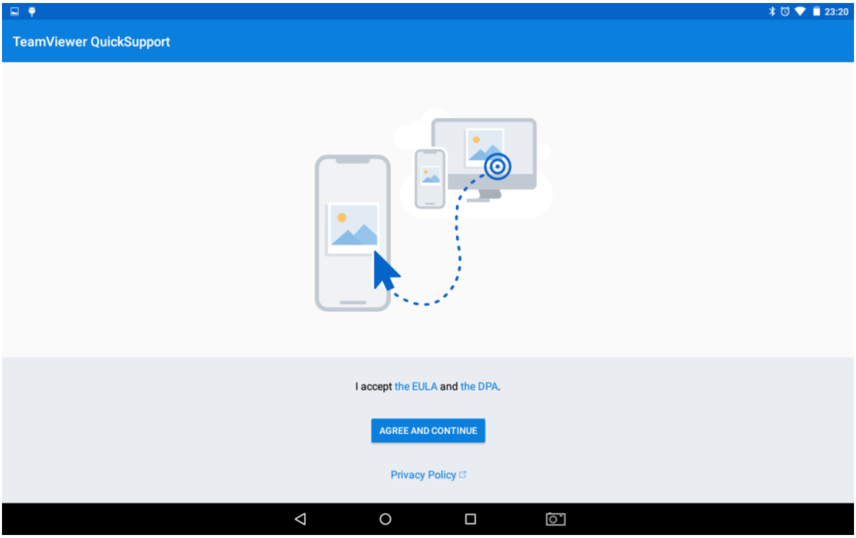
7 Remote Diagnosis

Through this function, you can receive remote service and support from after-sales technicians to assist you with vehicle model diagnosis.

➢ How to Receive Technical Support

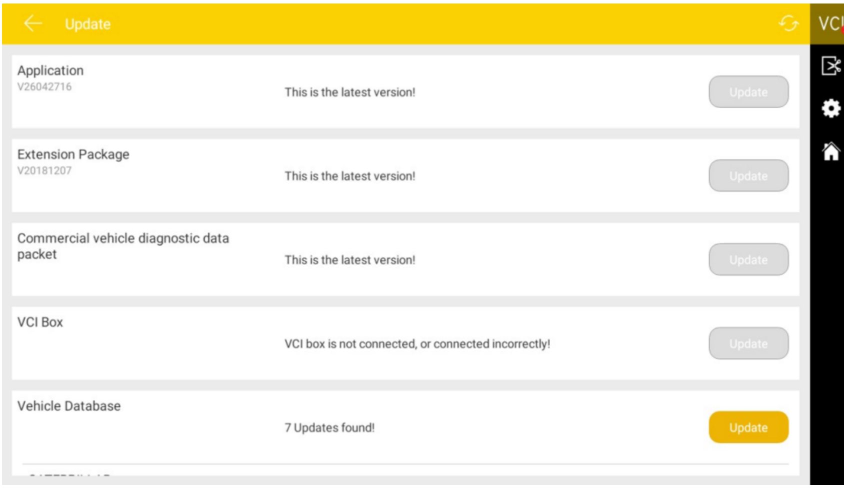
1) Turn on the power of the OHW808 host;
2) In the main function menu, select [Remote Diagnosis] to open the Team Viewer interface, and the system will automatically generate and display the device ID; **(Please ensure the device is connected to the internet.)**
3) Inform after-sales team of your device ID number and wait for them to send a remote control request;
4) After the system receives the request, choose [Allow] in the pop-up window to

accept, or choose [Reject] to decline.



8 Upgrade

By connecting the host device to the internet, you can upgrade the diagnostic software to promptly enhance the product's functionality. In the main function menu, open [Upgrade], and the system will automatically search for the latest update programs. As shown in the figure below, click [Upgrade] to update the vehicle models and other applications to the latest versions.

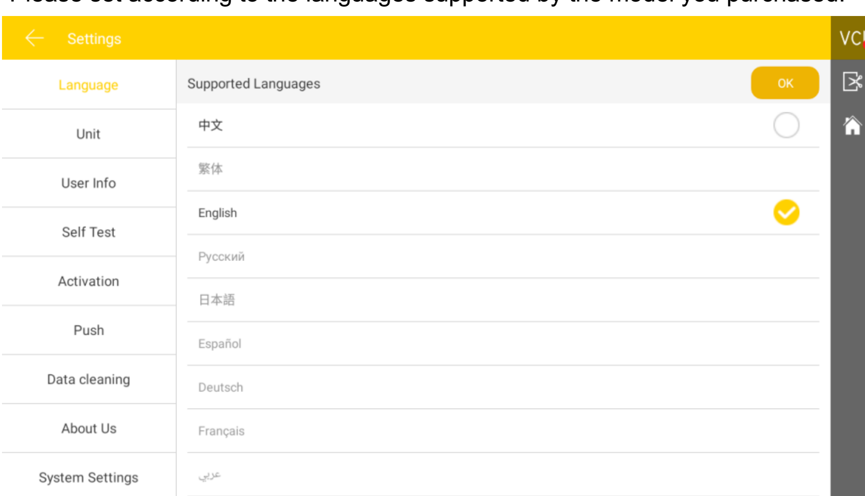


9 Settings

In the main function menu, select [Settings] to open the settings interface, where you can adjust the following system settings:

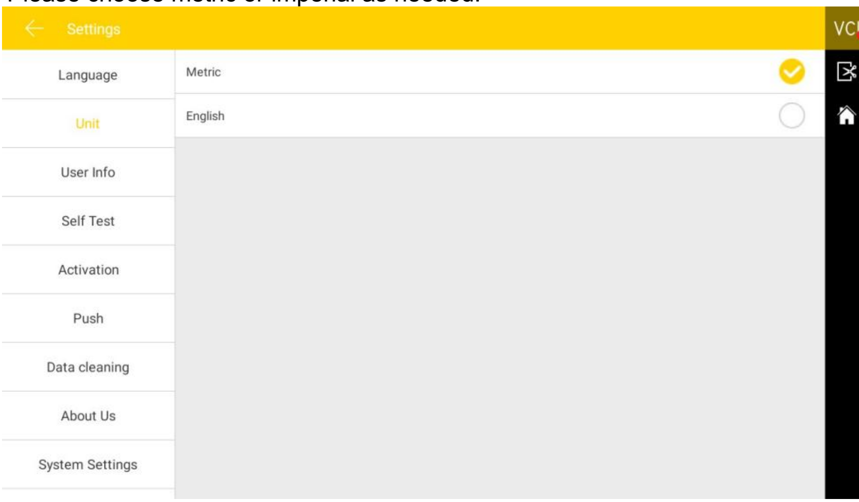
9.1 Language Settings

The OHW808 fault diagnostic instrument supports multiple language settings. Please set according to the languages supported by the model you purchased.



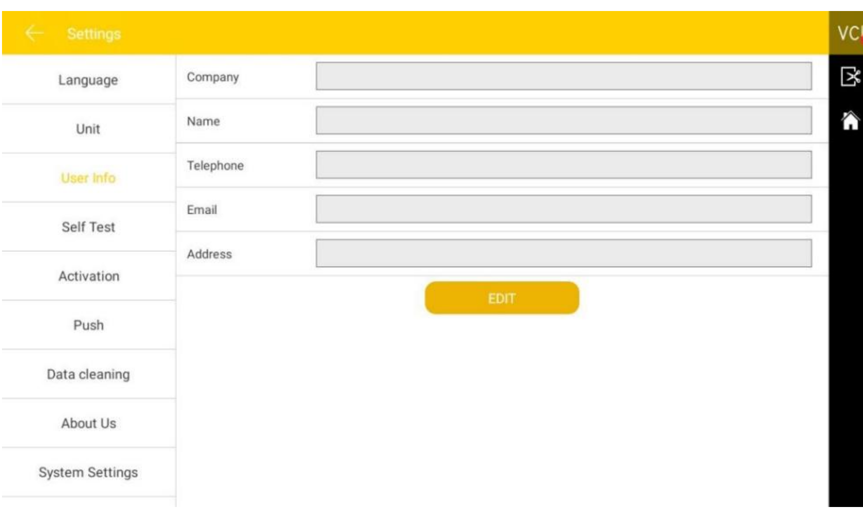
9.2 Units

This option allows you to set the data stream units in the diagnostic software. Please choose metric or imperial as needed.



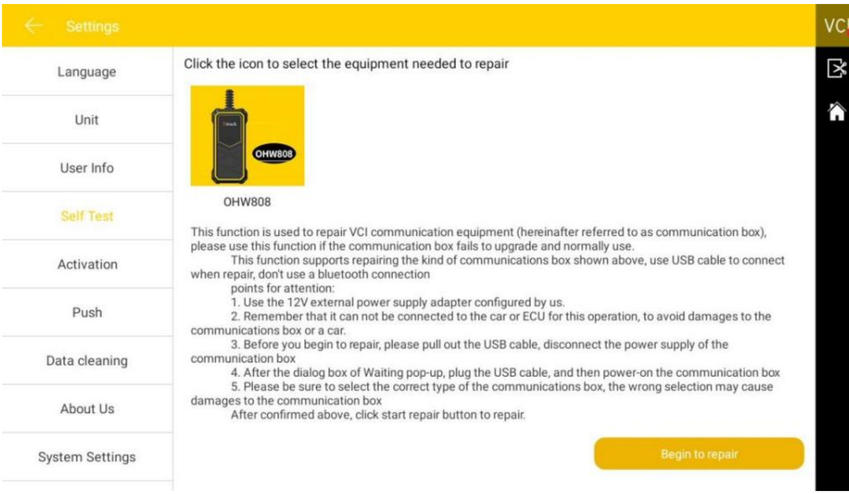
9.3 User Information

Set your personal information: name, phone number, email, address, etc.



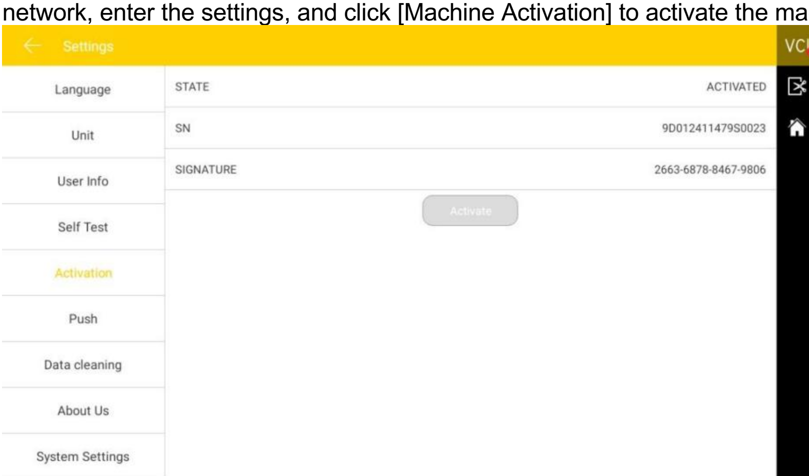
9.4 Machine Self-Test

Connect the device according to the screen illustration, and click [Begin to repair] to check for open circuits or short circuits in the main test leads and OBD-II connector, and determine their condition.



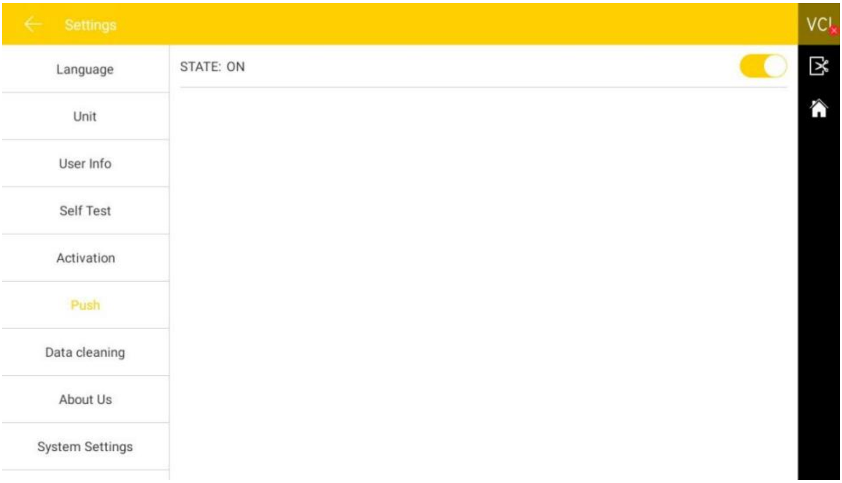
9.5 Machine Activation

OHW808 products are activated by number of uses when they leave the factory. After you turn on the machine and click on the vehicle model, it will prompt: "You are using a trial version, and you have ** remaining usage opportunities." Connect to the network, enter the settings, and click [Machine Activation] to activate the machine.



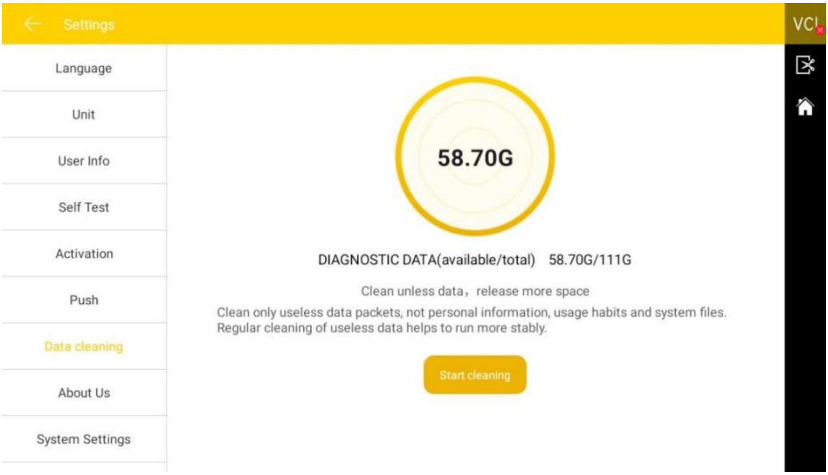
9.6 Message Push

Through the "Message Push" function, the OHW808 host can receive regular online messages from the server, such as system update notifications or other service message notifications. It is recommended that you always keep this function enabled so that you can receive the latest update services in a timely manner.



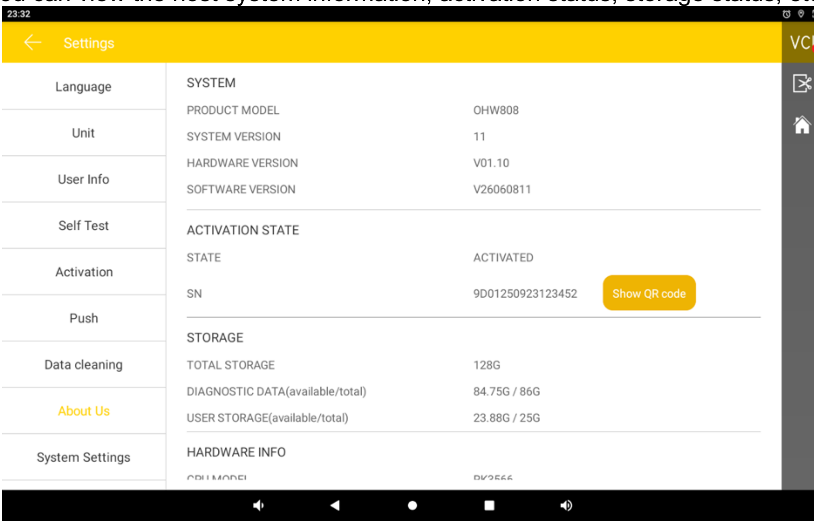
9.7 Data Cleanup

One-click cleanup of useless data to free up more space. Regular cleanup helps the system run more stably.



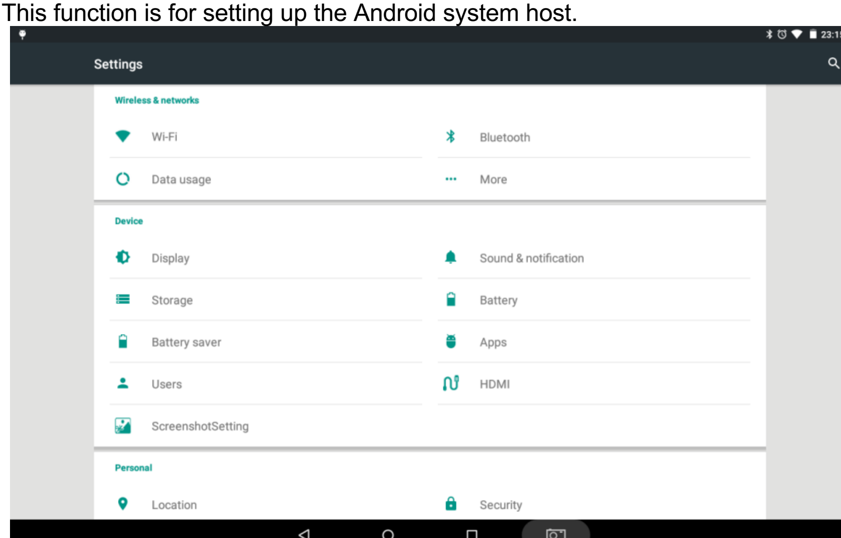
9.8 About Us

You can view the host system information, activation status, storage status, etc.



9.9 System Settings

This function is for setting up the Android system host.



Storage environment for care and maintenance

Storage Environment

1) **When not in use, the OHW808 automotive fault computer diagnostic instrument should be stored in a flat, dry, and temperature-appropriate place;**
2) **Do not place the OHW808 automotive fault computer diagnostic instrument in direct sunlight or near heating devices;**
3) **Please do not place the OHW808 automotive fault computer diagnostic instrument within the range of a magnetic field;**
4) **Do not place it in areas prone to smoke corrosion or where water or oil may splash;**
5) **Do not place it in areas prone to vibration, dust, humidity, or high temperatures;**

Host Protection

1) **Try to hold and place the host gently to avoid impact;**
2) **This machine must not be disassembled without authorization;**
3) **When connecting test leads and automotive diagnostics, be careful when plugging and unplugging. Tighten the screws during use to avoid damaging the interface during movement. After use, please loosen the screws first, then unplug the main test lead to avoid damaging the diagnostic interface;**
4) **After using the instrument, put the test leads and connectors and other accessories back into the box to avoid loss.**

Screen Maintenance

1) **The screen surface will attract dust due to static electricity. When the host is dirty, please cut off the power first, and use a soft cleaning cloth and neutral cleaner to clean it;**
2) **Do not wipe the dust with your fingers to avoid leaving fingerprints; do not use chemical cleaners to wipe the screen;**
3) **Do not place the product next to electrical equipment that generates**

electromagnetic wave interference to avoid affecting the display effect;

4) **Do not expose the screen to direct sunlight or ultraviolet light for a long time to avoid affecting the screen life.**

Precautions

1) **Prohibit frequent power on and off, sudden power outages, unstable power supply, and other abnormal human operations;**
2) **When not in use, please unplug the power plug to avoid aging of electrical components due to long-term power supply;**
3) **Avoid using in an environment with volatile acidic or alkaline chemicals to prevent corrosion of the host hardware;**
4) **Do not place any items on the display screen to avoid damaging the touch screen or internal components due to heavy pressure;**
5) **If the device is not used for a long period, power it on periodically to prevent moisture damage.**

Warranty Terms

Dear user, thank you for choosing our company's product. To better utilize this product, we recommend that you maintain it properly and follow the instructions in the user manual each time you use it. If your usage meets these requirements, the product will be able to provide you with longer-term high-quality service.

I. Under the following terms and conditions, and provided that you have activated the machine or registered as a user and completed your true information, if the product has defects in materials or workmanship, this company will provide product warranty services.
II. You confirm that you have read these product warranty terms carefully; otherwise, This company will regard the information you have registered and completed on the official website as your agreement and acceptance of these warranty terms.
III. Your product must be purchased from a product dealer authorized by this company. For products purchased through abnormal channels, the

purchaser shall bear the product repair service costs.

IV. The following items of the product are considered consumables: product user manual, test main line, connectors, inner and outer packaging boxes, promotional gifts, etc. Consumable items are not covered by the warranty.

V. From the date of purchase (subject to the valid purchase receipt and valid warranty card of the product), if the product experiences a performance failure due to non-human factors, within one month, you may choose to have the product repaired or to replace it with the same model. Within one year, you can enjoy free warranty services for the host and the power adapter.

VI. If your product is in any of the following situations, it will not be entitled to free warranty services:

1) Failures, defects, or flaws not caused by the product quality of this company: including your failure to use the product in accordance with the product manual, improper operation of the product, impact, drop, self-disassembly, connection of inappropriate accessories, improper transportation or storage of the product, resulting in compression damage, infiltration of liquids or food causing corrosion and rust, etc.;

2) Natural wear and tear of the product: including but not limited to the shell, buttons, touch screen, accessories, etc.;

3) The product serial number on the host does not match the serial number on the warranty card; the product quality inspection label or barcode has been removed, altered, or damaged;

4) Disassembly, repair, and modification not recognized by this company.

VII. If a product has quality problems or failures during the warranty period, you may take the following measures:

1) You may conduct a self-inspection of the product according to the product help information. If there are no hardware quality issues, you may attempt to upgrade the product's program;

2) You may contact the after-sales service center of this company to obtain correct service information;

3) After receiving a valid response from the company, you must send the product to the company's designated address (see the detailed address on this company official website) for repair and maintenance. Otherwise, your product may not be repaired and maintained in a timely manner, and you will bear the consequences if it is lost.

VIII. During the warranty service process, you will bear the costs related to the product being delivered or sent to the designated location of this company, including packaging, transportation, shipping, insurance, and other costs of the product.

IX. The free warranty service you enjoy under these warranty terms is the only measure for losses caused by product defects during the warranty period. this company shall not be liable for any direct or indirect losses you may suffer.

X. All warranty information, product functions, and changes to specifications will be published on the latest promotional materials and website of this company without further notice.

Certification

This product has been strictly inspected as qualified products and met the company standards.

Product name	Automotive Diagnostic System
Product serial number	
Date of production	
Inspector	

Warranty card

Product name	Automotive Diagnostic System
Product serial number	
Purchase date	

Company name: ______________________

User address: ______________________

Contact person: ______________________

Contact number: ______________________